SRI SHAKTHI

INSTITUTE OF ENGINEERING AND TECHNOLOGY

COIMBATORE – 62

Autonomous Institution, Accredited by NAAC with "A" Grade



21CS521 DATA SCIENCE AND ANALYTICS LABORATORY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY- COIMBATORE – 62

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

21CS521– DATA SCIENCE AND ANALYTICS

LABORATORY RECORD

NAME……..

ROLL NO : …….

CLASS : ……..…….

BRANCH:……..……...

ACADEMIC YEAR:……..

BATCH:……..……SEMESTER:……

Certified and Bonafide record of work done by…….……..

Place:

Date:

Staff In-Charge **Head of the Department**

University Register Number …….....

Submitted for the University Practical Examination held on……

INTERNAL EXAMINER **EXTERNAL EXAMINER**

INDEX

Exp.No	DATE	NAME OF THE EXPRIEMENT	MARKS	SIGNATURE
1		Download, install and explore the features of NumPy, SciPy, Jupyter, Statsmodels and Pandas packages.		
2		Pre-process the dataset to remove missing values, outliers and to transform and Discretize the dataset		
3		Implement the classification techniques: Decision Tree and Naïve Bayes algorithms and compare its performance.		
4(a)		Implement K-Means clustering algorithm		
4(b)		Implement K-Medoids clustering algorithms		
5		Implement Apriori algorithm to derive association rules.		
6		Classify the input dataset using Support Vector Machine (SVM)		
7		Demonstrate clustering approach using Kohonen's self-organizing map		
8		Implement Principal Component Analysis (PCA) technique to reduce the dimensionality of the dataset		

ADDITIONAL EXPERIMENTS

Exp.No	DATE	NAME OF THE EXPRIEMENT	MARKS	SIGNATURE
A1.1		Implementation of Neuralnetwork using Tensorflow for Iris Flower Classification		
A1.2		Student Pass/Fail Prediction using Tensorflow Binary Classification		
A1.3		Binary Classification Using Dropout and Early Stopping in Tensorflow		
A2		Dimensionality Reductio using LocallyLinearEmbedding(LLE)		
A3		Dimensionality Reduction using Isomap		
A4		Sale Performance Analysis using PowerBI		
A5		Discount & Profit Analysis using Tableau		